

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR

Odd Semester Mid-Term Examination, 2024-25

Course Code: CSC302

Course Name: Digital Logic Design

Full Marks: 25

Time: 90 Minutes

Answer all the questions from 1 to 7. PART-A

Question No.	Body of the Question	Marks	Mapped CO
1	State the inputs of the X1, X2, X3 to get the following circuit implements a two input AND gate using two 2-1 multiplexers.	2	CO1
2	Simplify the following Boolean function to product-of-sums form: $F(A, B, C, D) = \prod(1, 3, 5, 7, 13, 15)$	2	CO1
3	Find the Boolean function for the following K-map.	2	CO1
4	Construct a JK flip-flop, using a D flip-flop, a two-to-one-line multiplexer, and an inverter.	3	CO2
5	Using Boolean identities, reduce the given Boolean expression: $F(X, Y, Z) = X'Y + YZ' + YZ + XY'Z'$	2	CO2
6	An 8 X 1 multiplexer has inputs A, B, and C connected to the selection inputs S2, S1, and S0 respectively. The data inputs I0 through I7 are as follows: I1=I2=I7=0; I3=I5=1; I4=I0=D; and I6 = D'. Determine the Boolean function that the multiplexer implements.	2	CO1
7	Consider the minterm list form of a Boolean Function F given below: $F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$ Here m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is _____.	2	

PART-B

Answer Any **Two(2)** only from the Questions given below (8-11)

5X2=10

8

Let $G(x_1, x_2, x_3, x_4)$ and $F(x_1, x_2, x_3)$ be Boolean functions given by the K-maps shown below.

(a) Give a minimal SOP representation of F and a minimal POS representation of G . A minimal representation uses as few minterms/ maxterms as possible, and each minterm/maxterm uses as few literals as possible.

$x_1, x_2 \rightarrow$ $x_3, x_4 \downarrow$	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	0	1	0	1
10	1	1	0	1

$x_1, x_2 \rightarrow$ $x_3 \downarrow$	00	01	11	10
0	1	1	0	0
1	0	1	1	1

5

CO2

(b) Implement a full adder with two 4 X 1 multiplexers.

9

(a) Define the carry propagate and carry generate as $P_i = A_i + B_i$, $G_i = A_i B_i$ respectively. Show that the output carry and output sum of a full adder becomes $C_{i+1} = (C_i' G_i' + P_i')$ and $S = (P_i G_i') \oplus C_i$

(b) A PN flip-flop has four operations, clear to 0, no change, complement, and set to 1, when inputs P and N are 00, 01, 10, and 11, respectively. Tabulate the characteristic table. Derive the characteristic equation. Tabulate the excitation table. Show how the PN flip-flop can be converted to a D flip-flop.

5

CO2,
CO3

10

Design a combinational circuit with three inputs, x , y , and z , and three outputs, A , B , and C . When the binary input is 0, 1, 2, or 3, the binary output is two greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is three less than the input.

5

CO2

11

(a) A combinational circuit is specified by the following three Boolean functions:

$$F_1(A, B, C) = \sum(3, 5, 6), F_2(A, B, C) = \sum(1, 4), F_3(A, B, C) = \sum(2, 3, 5, 6, 7)$$

Implement the circuit with a decoder constructed with NAND gates and NAND or AND gates connected to the decoder outputs. Use a block diagram for the decoder. Minimize the number of inputs in the external gates.

(b) If X and Y are Boolean variables, which one of the following is the equivalent of $X \oplus Y \oplus XY$?

- (i) $X + Y'$ (ii) $X + Y$ (iii) 0 (iv) 1

5

CO1,
CO2